

The solver: building, solving, interpreting



Outline

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The general commands



Introduction: The general commands

- ▶ This chapter is the **toolbox** of the laboratory
- ▶ How to build a model with gurobipy, how to run it, how to retrieve the solution
- ▶ And — the most important part — how to **interpret the output**
- ▶ The examples use the same LPs as the background chapter: together they provide the basis for all the models of the following chapters

Installation and licence

```
python3 -m pip install gurobipy
```

- ▶ The package includes a **demonstration licence** (up to 2000 variables and 2000 constraints), sufficient for almost all the models in these lecture notes
- ▶ Full and free **academic licence**: register at <https://portal.gurobi.com> with your institutional email address
- ▶ Request a *Named-User Academic License* and run the command `grbgetkey` indicated by the portal (on the university network or VPN)



The five steps to build a model

- ▶ Let us take up again the 2×2 LP of the background chapter:

$$\max \quad 30x_1 + 50x_2$$

$$x_1 + 3x_2 \leq 90 \quad (\text{constraint 1})$$

$$2x_1 + x_2 \leq 80 \quad (\text{constraint 2})$$

$$x_1, x_2 \geq 0$$

- ▶ We build it in gurobipy in five steps: container, variables, constraints, objective, writing to file

The five steps (1/2)

```
import gurobipy as gp
from gurobipy import GRB

# Step 1: the model container
m = gp.Model("lp_2x2")

# Step 2: the decision variables
#           (default: continuous, >= 0)
x1 = m.addVar(name="x1")
x2 = m.addVar(name="x2")
```

The five steps (2/2)

```
# Step 3: the constraints (keep the object
#           to read the dual later)
v1 = m.addConstr(x1 + 3*x2 <= 90, name="vincolo1")
v2 = m.addConstr(2*x1 + x2 <= 80, name="vincolo2")

# Step 4: the objective function
m.setObjective(30*x1 + 50*x2, GRB.MAXIMIZE)

# Step 5 (debug): write the model
#           in readable format
m.write("lp_2x2.lp")
```

The three golden rules of gurobipy (1/2)

1. Indexed variables with addVars:

```
x = m.addVars(I, T, name="x")
```

- ▶ creates the family $x[i, t]$ over the index sets I and T
- ▶ the sums are written `gp.quicksum(...)` or `x.sum(i, "*")`

2. Free variables (which may be negative, such as the intercept b of the SVM or the threshold η of the CVaR):

```
m.addVar(lb=-GRB.INFINITY)
```

The three golden rules of gurobipy (2/2)

- ▶ Forgetting lb is the **most frequent mistake** in the laboratory: the default is ≥ 0 !

3. Families of constraints with a generator:

```
m.addConstrs((x.sum("*", t) <= b[t] for t in T),  
              name="cap")
```

Running the model

```
m.Params.OutputFlag = 0    # 0 = silent, 1 = log
m.Params.TimeLimit   = 60  # seconds
m.Params.NonConvex   = 2   # only for non-convex QPs
                        # (e.g. pricing p*q)
m.optimize()
```

- ▶ Parameters are set *before* optimize()
- ▶ NonConvex is only needed for non-convex QPs

Reading the log

The solver log, when active, must be read at three points:

```
Optimize a model with 2 rows, 2 columns and 4 nonzeros  
Coefficient statistics: Matrix range [1e+00, 3e+00]  
Optimal objective 1.900000000e+03
```

- ▶ line 1: is the **size** the expected one?
- ▶ line 2: are the data **well scaled**?
- ▶ line 3: **status** and optimal value

If the coefficient ranges span from 10^{-6} to 10^9 the model is badly scaled: change the units of measurement (thousands of euros, MWh, ...) before looking for other errors

Retrieving the solution

Always check the status before reading any number:

```
if m.Status == GRB.OPTIMAL:
    print(m.ObjVal, x1.X, x2.X)
elif m.Status == GRB.INFEASIBLE:
    # which constraints contradict each other
    m.computeIIS(); m.write("conflitto.ilp")
elif m.Status == GRB.UNBOUNDED:
    # almost always a constraint or a bound is missing
    pass
```

The main attributes

Attribute	Object	Meaning
Status	model	OPTIMAL=2, INFEASIBLE=3, UNBOUNDED=5, TIME_LIMIT=9
ObjVal	model	optimal value of the objective
X	variable	value of the variable at the optimum
Slack	constraint	slack: 0 $\Rightarrow$ active constraint

- The attributes specific to LPs (shadow prices, reduced costs, validity ranges) are in the section that follows

Summary

Building:

- ▶ Five steps: `Model`, `addVar(s)`, `addConstr(s)`, `setObjective`, `write`
- ▶ Beware of the default ≥ 0 : free variables need `lb=-GRB.INFINITY`

Solving and reading:

- ▶ In the log: size, scaling of the coefficients, status
- ▶ Before reading `ObjVal`, `X`, `Slack`: always check `Status`

The commands for linear models

Introduction: The commands for linear models

- ▶ For LPs the solver returns much more than the solution: **shadow prices**, **reduced costs** and **validity ranges**
- ▶ We see how to read them, with which sign, and a checklist to interpret the output
- ▶ We close with two complete readings: the 2×2 LP and the “all the cases” LP

The bridge with the background notation

The solver attributes are exactly the theoretical quantities of the background chapter:

Theory	Gurobi attribute	Object
shadow price $\tilde{\pi}_i$	Pi	constraint
reduced cost $\bar{c}_j$	RC	variable
slack $\bar{s}_i$	Slack	constraint
range $[b_i^{\min}, b_i^{\max}]$	SARHSLow–SARHSUp	constraint
range $[c_j^{\min}, c_j^{\max}]$	SAObjLow–SAObjUp	variable

The stored RHS

Observation 1: The RHS is not always the one you wrote

If a constraint contains a constant on the left-hand side, such as

$$\begin{aligned} &\text{quicksum}(x[v,t] \dots) \\ &\quad + \text{base}[t] \leq C \end{aligned}$$

Gurobi moves it into the right-hand side: the stored RHS is $C - \text{base}[t]$.

- ▶ Writing `constr.RHS = nuovaC` in fact imposes `quicksum(x) ≤ nuovaC`, which is something else
- ▶ The correct form is `constr.RHS = nuovaC - base[t]`
- ▶ This mistake actually happened while preparing the case study on electric vehicle charging

Interpreting the output: the checklist (1/2)

1. Is the status OPTIMAL? If not, stop and diagnose (compute IIS for infeasibility, the .lp file for unboundedness)
2. Does the optimal value have the expected order of magnitude?
3. Which constraints are active (Slack = 0)? Are they the expected ones?
4. How large are the shadow prices? Which resource is worth expanding first? (Signs: in a minimization problem, a constraint $\leq$ has $P_i \leq 0$)

Interpreting the output: the checklist (2/2)

5. Are there variables at zero? Read their reduced cost (RC): it says by how much the objective coefficient must improve for them to be worth activating
6. Is the solution stable? Re-run with data perturbed by 5%
7. Translate everything into a three-line managerial recommendation



Example: complete reading of the 2×2 (1/3)

Running the code of the five steps:

```
Status: 2 (OPTIMAL)          ObjVal: 1900.0
x1 = 30.0    RC = 0.0    SAObj = [16.7, 100.0]
x2 = 20.0    RC = 0.0    SAObj = [15.0, 90.0]
vincolo1: Slack=0.0    Pi=14.0    SARHS=[40.0, 240.0]
vincolo2: Slack=0.0    Pi=8.0    SARHS=[30.0, 180.0]
```

Example: complete reading of the 2×2 (2/3)

- ▶ Both constraints are **active** (binding)
- ▶ One extra unit of b_1 is worth 14, and the valuation remains valid as long as b_1 stays in $[40, 240]$; beyond that, everything must be recomputed
- ▶ If only one right-hand side could be increased, better b_1 ($14 > 8$ per unit, at equal cost)

Example: complete reading of the 2×2 (3/3)

- ▶ The reduced costs (attribute $x_1.RC$) are null because both variables are basic
- ▶ Careful: there may also exist a variable *basic at value zero* (degeneracy), so $RC = 0$ alone does not say that the variable is used
- ▶ Adding the third variable of the background chapter the solver would give $x_3 = 0$ with $RC = -2$
- ▶ Its coefficient must rise by at least 2 (that is, to 22) for it to enter the solution, with range $SAObj = (-\infty, 22]$

In the general case: shadow prices and sign (1/2)

- ▶ **Shadow prices** (attribute P_i , one per constraint): always the “derivative of the optimal value with respect to the right-hand side”
- ▶ The sign is deduced in two moves:
 - ▶ increasing b_i **enlarges** the feasible region (constraint $\leq$) or **shrinks** it (constraint $\geq$)
 - ▶ a wider region can never worsen the optimum

In the general case: shadow prices and sign (2/2)

- ▶ *Maximization* problem: resource constraint $\leq \Rightarrow P_i \geq 0$ (more resource, more profit); constraint $\geq \Rightarrow P_i \leq 0$
- ▶ *Minimization* problem: demand constraint $\geq \Rightarrow P_i \geq 0$ (more demand, more cost); capacity constraint $\leq \Rightarrow P_i \leq 0$ (more capacity, less cost)
- ▶ *Equality* constraint $\Rightarrow$ any sign, as prescribed by the rules of duality
- ▶ Inactive constraint $\Rightarrow P_i = 0$; validity within the range $SARHS_{Low} - SARHS_{Up}$

In the general case: reduced costs

Reduced costs (attribute RC, one per variable):

- ▶ *basic* variable: $RC = 0$ — as a rule it lies strictly between its bounds, but it may also sit on a bound (*degenerate* basis)
- ▶ variable at its *lower bound* (typically at zero): $RC \geq 0$ in a minimization (by how much its cost must fall), $RC \leq 0$ in a maximization (by how much its margin must rise); validity in SAObjLow-SAObjUp
- ▶ variable at its *upper bound*: RC is the shadow price of the bound — in a minimum-cost flow a saturated arc has $RC < 0$: one extra unit of capacity saves $|RC|$

The complete loop

For *all* the constraints and variables (LP only):

```
for v in m.getConstrs():  
    print(v.ConstrName, v.Pi, v.SARHSLow, v.SARHSUp)  
for v in m.getVars():  
    print(v.VarName, v.X, v.RC, v.SAObjLow, v.SAObjUp)
```

Example: all the cases (1/5)

- ▶ We implement the “all the cases” LP of the background chapter: three constraint directions, three variable signs

$$\min \quad 5x_1 + 8x_2 - 9x_3$$

$$x_1 + x_2 \geq 30$$

$$x_1 + x_2 - x_3 = 100$$

$$x_1 - 2x_2 \leq -20$$

$$x_1 \geq 0, \quad x_2 \text{ free}, \quad x_3 \leq 0$$

- ▶ We read *everything* the solver returns

Example: all the cases (2/5)

```
m = gp.Model("tutti_i_casi")
x1 = m.addVar(name="x1") # x1 >= 0
x2 = m.addVar(lb=-GRB.INFINITY, name="x2") # x2 free
x3 = m.addVar(lb=-GRB.INFINITY, ub=0,
              name="x3") # x3 <= 0
v1 = m.addConstr(x1 + x2 >= 30, name="vincolo1")
v2 = m.addConstr(x1 + x2 - x3 == 100, name="vincolo2")
v3 = m.addConstr(x1 - 2*x2 <= -20, name="vincolo3")
m.setObjective(5*x1 + 8*x2 - 9*x3, GRB.MINIMIZE)
m.optimize()
```

Example: all the cases (3/5)

```
for v in m.getVars():  
    print(v.VarName, v.X, v.RC, v.SAObjLow, v.SAObjUp)  
for c in m.getConstrs():  
    print(c.ConstrName, c.Slack, c.Pi,  
          c.SARHSLow, c.SARHSUp)
```

```
Status: 2 (OPTIMAL)      ObjVal: 620.0  
x1: X = 60.0 RC = 0.0 SAObj = [-inf, 8.0]  
x2: X = 40.0 RC = 0.0 SAObj = [5.0, 17.0]  
x3: X = 0.0 RC = -3.0 SAObj = [-inf, -6.0]  
vincolo1: Slack=-70.0 Pi= 0.0 SARHS=[-inf, 100.0]  
vincolo2: Slack= 0.0 Pi= 6.0 SARHS=[30.0, inf]  
vincolo3: Slack= 0.0 Pi=-1.0 SARHS=[-200.0, inf]
```

Example: all the cases (4/5)

Here all the cases of the previous explanation appear:

- ▶ the **inactive** constraint with $P_i = 0$ (vincolo1, Slack $\neq 0$)
- ▶ the **equality** with a positive dual (+6)
- ▶ the $\leq$ in a **minimization** with a negative dual (-1)

Example: all the cases (5/5)

- ▶ two **basic** variables with $RC = 0$
- ▶ the variable x_3 stuck at its **upper bound** (0) with $RC = -3$ and profitability threshold $SAObjUp = -6$
- ▶ The verification by hand of each of these numbers, with the rules of duality and by perturbation, is in the “all the cases” example of the background chapter

Summary

LP attributes:

- ▶ P_i , RC, Slack and the ranges SARHS/SAObj are the theoretical quantities of the background chapter
- ▶ The stored RHS includes the constants moved from the left-hand side

Interpretation:

- ▶ The sign of P_i follows from the constraint direction and the optimization sense
- ▶ $RC = 0$ is not enough to say that a variable is used (degeneracy)
- ▶ Follow the checklist up to the managerial recommendation

The commands for nonlinear models

Introduction: The commands for nonlinear models

- ▶ A single solver for the whole laboratory
- ▶ Gurobi covers LP and QP natively (including non-convex ones, with `m.Params.NonConvex = 2`)
- ▶ From version 12 it also solves *globally* models with general nonlinear functions
- ▶ We see how to write these models, how to handle tolerances and scaling, and what to do when the duals are not there

Functional constraints and bilinear terms

- ▶ Nonlinear functions — logarithms, exponentials, powers — enter the model through **functional constraints**
- ▶ An auxiliary variable y is introduced and $y = f(x)$ is imposed
- ▶ Ratios such as $1/(\mu - \lambda)$ are instead reformulated with an auxiliary variable and a **bilinear constraint**:

$$w(\mu - \lambda) = 1$$

Functional constraints and bilinear terms: the code

```
# functions handled exactly (globally)  
m.Params.FuncNonlinear = 1  
z = m.addVar(lb=-GRB.INFINITY)  
# z = log(g); also Exp, Pow, Sin, ...  
m.addGenConstrLog(g, z)  
# bilinear terms: NonConvex = 2 is required  
m.addQConstr(w * v == 1)
```

Tolerances and scaling

Two practical devices, both used in the scripts of these lecture notes:

- ▶ **Tolerances:** for accurate marginal analyses (differences between two nearby optima) tighten MIPGap, FeasibilityTol and OptimalityTol to 10^{-9}
- ▶ **Scaling:** reformulate to avoid tiny quantities — for instance

$$q \leq A p^{-\varepsilon} \quad \text{is written} \quad q \cdot r \leq A \quad \text{with} \quad r = p^{\varepsilon}$$

which keeps the numbers in a healthy range

When the duals are not there

- ▶ For models with functional or bilinear constraints the solver **does not provide the duals**
- ▶ Shadow prices and multipliers are estimated *by perturbation*:
 1. increase the right-hand side by ε
 2. re-optimize
 3. take the incremental ratio
- ▶ It is the same protocol as in the background chapter

Why a single solver

- ▶ Same syntax, same interpretation checklist
- ▶ Above all: a *certified global* optimum even in non-convex problems
- ▶ With a local solver every result would have to be accompanied by the question “is it only a local optimum?”
- ▶ Convex problems with a purely quadratic or conic structure (constraints such as $d_x^2 + d_y^2 \leq d^2$) require nothing special

Exercise: shadow price by perturbation

- ▶ Implement the 2×2 LP
- ▶ Print P_i , Slack, SARHSLow/Up of both constraints
- ▶ Verify the shadow price of the first constraint by re-optimizing with $b_1 : 90 \rightarrow 91$

Solution

- ▶ The solver prints the values of the complete reading: $P_i = 14$ and 8, $Slack = 0$, ranges $[40, 240]$ and $[30, 180]$
- ▶ With $b_1 = 91$ both constraints remain active:

$$x_1 + 3x_2 = 91, \quad 2x_1 + x_2 = 80 \quad \Rightarrow \quad x_1 = 29.8, \quad x_2 = 20.4$$

- ▶ New optimum $30 \cdot 29.8 + 50 \cdot 20.4 = 1914$: the increase is $1914 - 1900 = 14 = P_i$ of constraint 1

Exercise: missing constraint and objective sense

- ▶ Remove the second constraint and observe the outcome
- ▶ Then change the objective to minimization with no other changes: why is the result $x = 0$?

Solution

- ▶ Without constraint 2 the problem is still bounded: the vertices are $(0, 0)$, $(90, 0)$, $(0, 30)$ with values 0, 2700, 1500
- ▶ Status OPTIMAL with $x_1 = 90$, $x_2 = 0$, $ObjVal = 2700$
- ▶ In minimization the coefficients 30 and 50 are positive and the variables are ≥ 0 (default!): the objective is ≥ 0 and equals 0 at $x = 0$, which is feasible
- ▶ Moral: without constraints that “push” (demand to serve) a minimum-cost problem does nothing

Exercise: diagnosing infeasibility

- ▶ Deliberately build an infeasible model, with two incompatible constraints:

$$x_1 + x_2 \geq 10, \quad x_1 + x_2 \leq 8$$

- ▶ Use `computeIIS` to identify the conflict

Solution

- ▶ `m.Status` is INFEASIBLE (3): no number of the solution should be read
- ▶ `m.computeIIS()` computes an *irreducible* infeasible subset of constraints: here the two constraints on $x_1 + x_2$
- ▶ `m.write("conflitto.ilp")` writes them to file; for each constraint the attribute `IISConstr` is 1 if it belongs to the conflict

Exercise: a functional constraint

- ▶ Solve with Gurobi's functional constraints the one-variable problem

$$\max 5 \log(1 + x) - x, \quad x \in [0, 10]$$

- ▶ Compare with the analytical solution:

$$\frac{5}{1+x} - 1 = 0 \quad \Rightarrow \quad x = 4$$

Solution

- ▶ Variables: $x \in [0, 10]$, $g = 1 + x$ and z free, with the functional constraint $z = \log(g)$ (`addGenConstrLog`)
- ▶ Objective linear in the new variables: $\max 5z - x$
- ▶ The solver returns $x = 4$ and optimal value $5 \log 5 - 4 \approx 4.047$, as the analytical solution

Summary

Nonlinear models:

- ▶ Nonlinear functions through functional constraints $y = f(x)$; ratios through bilinear constraints (`NonConvex = 2`)
- ▶ Tight tolerances for marginal analyses; reformulate for good scaling

Duals and solver:

- ▶ Without duals, shadow prices are estimated by perturbation
- ▶ A single solver: same syntax and certified global optimum

Final summary

- ▶ **Building:** five steps in gurobipy, beware of the default ≥ 0 of the variables
- ▶ **Solving:** read the log (size, scaling, status) and always check Status
- ▶ **Interpreting:** Pi, RC, Slack and ranges, with the right signs and the checklist
- ▶ **Nonlinear:** functional and bilinear constraints, duals estimated by perturbation